Erik Hill

AI Evaluation & Testing Engineer

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SUMMARY

Self-taught AI evaluation and testing engineer building deterministic, replayable systems for measuring coding agents and LLM applications.

Merged upstream at promptfoo (by its co-founder), GitLab and TeaVM; an arXiv preprint; **30+ public repos, three PyPI packages, one npm package**; a game shipped to Google Play. I build **verifiable-evaluation systems** whose claims are checked rather than asserted: **a public registry of 12 evaluation claims my verifier accepted, each hash-pinned to a commit and replayed weekly in CI** (11 of 12 reproduced in the 2026-11-02 replay): deterministic, with failures published by design.

EXPERIENCE

AI/ML Engineer · Epix Devo LLC · *Contract · Oct 2026 to Present*

- Contract AI/ML engineering against approved requirements: model and framework evaluation, proof-of-concept prototyping, and integration with APIs, data stores and application backends.

AI Evaluation & Testing Engineer · Self-employed · *Remote · Mar 2026 to Present*

- Published **evalmut** (mutation testing for LLM eval graders) to **PyPI**, Aug 2026; promptfoo merges below.
- Built a **multi-agent LLM coding harness**: a plan, build, review, ship loop with **per-role model routing**, a fresh-context LLM reviewer, an asynchronous **evaluation** role and a human on every consequential step; gates written as **Claude Code hooks**, later ported to pi-gates; property-based invariants over a native game engine and adb-driven on-device checks. Write-up (source private): exposure-watch/docs/HARNESS.md
- Operate **model-drift**, a public **LLM regression tracker**, in **Python** on a daily **GitHub Actions** schedule: it calls the **OpenAI, Anthropic, Gemini, xAI and Groq APIs** over REST, grades a frozen 35-task suite with deterministic checks, scores accuracy, latency, verbosity, reliability and refusal rate, and since Oct 2026 writes every run to **PostgreSQL**. Run daily since 2026-07-19 (one failed day); now **17 models / 5 providers**.
- Diagnosed an apparent **66-point overnight drop** (69% to 3%) as **HTTP 429 rate limiting** on Groq's shared 30-requests-per-minute tier, not model drift; shipped Retry-After-aware retries with exponential backoff, a per-host throttle and a reliability floor (runs under 50% call success leave the accuracy chart), and stopped failed calls scoring as wrong answers.

Indie Game Developer · SeraphLight Studios (self-employed) · *Remote · Nov 2025 to Present*

- **Shipped Tap Dodge Rush to Google Play** (Kotlin / Java, Android Views + Canvas, AdMob). A second, larger game (React / Capacitor) was in active development through August 2026.

Line Cook to Kitchen Manager / Shift Lead · Big Island Brewhaus & other high-volume kitchens · *2016 to 2025*

- Led shifts and trained staff; owned prep, inventory, and food safety under sustained pressure.

SKILLS

AI/ML engineering: LLMs · Generative AI (LLM application development) · OpenAI, Anthropic and Gemini APIs · Claude Code · Machine learning (LoRA fine-tuning with MLX)
Evals & AI: Deterministic grading · Mutation testing · Differential testing · Property-based testing · Regression gates in CI · Minimum-detectable-change floors · RAG (BM25, hybrid retrieval, nDCG@10) · Tool / function calling & guardrails · Safety (OWASP LLM Top 10, human-in-the-loop) · Prompt engineering · Context engineering · Model Context Protocol (MCP) · LangGraph · Multi-agent orchestration
Backend & data: Python · FastAPI · Pydantic · SQL · PostgreSQL (Neon) · SQLAlchemy 2.0 · pgvector · REST
Frontend / mobile: TypeScript · JavaScript · React · Next.js · Java · Capacitor · Android · WebAssembly · TeaVM
Ship & verify: pytest · Vitest · JUnit · jqwik · Playwright · Docker / Compose · CI/CD (GitHub Actions) · Prometheus metrics · Structured logging · PyPI / npm packaging · Cloudflare Workers · Test automation · QA for LLM systems

OPEN-SOURCE CONTRIBUTIONS & EXTERNAL REVIEW

- **Two pull requests merged into promptfoo** (26k stars), both merged by its co-founder: #11278 (Sep 2026) stopped the BLEU assertion scoring a blank reference 9.99e-8 instead of 0 (my fix and test; the co-founder condensed the fix and rewrote the tests; released in 0.124.0); #10256 (Oct 2026) added the `rouge-l` / `rouge-s` assertion types (my four commits, which the co-founder integrated and refactored).
- **Three merge requests merged into GitLab**: two reconciling the CVSS fields of three CVE records with their vectors (CVE-2023-5009 stored 9.6 CRITICAL while the record's own vector string computes to 8.2 HIGH), and one correcting the security handbook's disclosure window from 30 days to 90.

• **Outside review of vac-protocol:** a reviewer's multi-agent audit falsified my own verifier, cheapest forgery **four bytes**. Fix merged, then I found the same defect class **four more times** (arXiv, below). As **sole maintainer** I merged four of his pull requests and answered his four specification issues (protocol v0.2); on two I showed his proposed fix would not close the hole and shipped a wider one.

• **Outside critique, credited in the repo:** Bartłomiej Nawara's missing-not-at-random critique of **model-drift**'s 429 handling shaped its **reliability floor**.

• **Closed a bug report open in TeaVM since 2018** that its own suite could not catch: a **one-character error** in its GregorianCalendar emulation, found by differential-testing the browser port against the JVM. The patch re-enables the four commented-out assertions covering that line; merged same day by the maintainer.

SELECTED PROJECTS, all public at github.com/egnaro9

• **exposure-watch**: **Does an AI agent beat plain code at known-exploited-vulnerability triage?** Pre-registered, 57 held-out pairs: a **tie on correct answers**, 47 to 45, p = 0.69, but the agent pipeline with a skeptic sent its 10 unsure cases to a person and got **0 wrong**, vs. 12 wrong from code plus a keyword rule. Measured with planted errors on the first 22-pair set, the **LLM skeptic caught 22 of 22 opposite-verdict flips** that the deterministic quote verifier passes by design, and rejected 2 of 22 correct answers.

• **agent-certlab**: **Two real coding agents certified, every verdict regradeable from artifacts**. Three **seeded-defect task families** with provable answer keys, a **precondition prover** (clean passes, every seed fails; it refused one of my own defects), and **calibration agents** the harness must separate before certifying.

• **harness-builder**: **More scaffolding scored worse, for 22x the cost**: one drafter 95%, planner plus drafter 90%, four-call panel with judge 80% on one 20-task suite, zero errors in 60 runs. The panel beat the single drafter on **0 of 20 tasks**, yet with 17 ties the tool reported **"this suite cannot decide"**.

• **reference-fleet**: **An answer key for eval suites**: six scripted reference models each broken **one documented way at an exactly reproducible seeded rate**; a naive suite detects **1 of 6**. A **LoRA-trained native defect** showed training's rate control is lossy: mixture 0.507 → **0.20 greedy, 0.31 at temp 1.0**.

• **evalmut**: **Mutation testing for eval graders** (built on its **gradecore** engine): inject known defects, report which checks stayed green. **No LLM judge**; every operator names its origin. Found **2 blind spots and 2 coverage gaps in gradecore's own graders** and **8 in my ports of a widely-used framework's assertions**; separately fixed **three other fail-open graders** in gradecore 0.10.3. *319 tests pass, 6 skipped.*

• **deadcheck**: **does an eval rubric catch what it claims?** Run on **three rubrics I did not write**: OpenAI evals' `Match` accepts "10" for ideal "1"; a promptfoo example's refusal test passes a lowercased refusal-then-compliance reply. Seven probe rounds found **23 defects in deadcheck itself under a passing suite**. *185 tests.*

• **vac-protocol**: **Verifiable Agent Claims**: evidence-bundle format + **offline verifier** behind the registry above, with a **public record of it rejecting my own evidence**. Plus **vac-gate**: no verified contract, no green check.

• **deadgate**: **a linter for CI checks that cannot fail** in GitHub Actions workflows (**PyPI**). Found an unguarded required status check in **heygen-com/hyperframes**, fixed by my merged PR #5182. *228 tests.*

• **rag-eval-lab**: deterministic RAG **eval harness** that catches a planted hallucination in CI; from-scratch **BM25 matching the published SciFact baseline (0.664 vs 0.665)**. *60 tests.*

• **eval-history**: model-drift's **run store** on Neon **PostgreSQL**, written daily since Oct 2026 and archived daily as JSON: **SQLAlchemy 2.0**, **Alembic** migrations diffed against the models in CI, and a **FastAPI** API with Prometheus /metrics and JSON logs (tested, no longer hosted). *67 tests, coverage gate 88%.*

• **llm-gateway**: a **FastAPI LLM gateway** that puts OpenAI, Anthropic and an offline mock behind one OpenAI-style endpoint: API-key auth, per-key **token-bucket rate limits**, a per-tenant cache, retries, SSE streaming, estimated cost accounting, Docker config. agent-graph can route through it. *29 tests, run against the mock.*

• **agent-graph**: a **LangGraph ReAct agent** with **unit-testable** multi-step tool use: a swappable mock/LLM policy runs it deterministically without a key; a **safe AST calculator** (no `eval`); a max-step guard. *67 tests.*

• **Also public: tapdodge-engine** (JVM-versus-browser differential check in CI) · **mcp-tools** (MCP server, no SDK) · **pi-gates** / **pi-eval** (fail-closed gates and deterministic grading for the pi coding agent) · **llm-wire-stub** (npm) · **readme-test-count** (fails CI when a README's test count disagrees with the suite)

PUBLICATIONS

"Four Ways to Forge a Bundle My Own Verifier Calls Clean: Refusal-Site Mutation Testing of an Evidence-Bundle Verifier." Erik Hill. **arXiv preprint, 2026**. arXiv:2608.26183
Drift Notes: erikhill.dev/notes

EDUCATION

Clark College, 2007 to 2009 (some college, no degree).